

Data Collection and Preprocessing Phase

Date	06 September 2026
Team ID	
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The data consists of 345,408 rows and 10 columns that correspond to different agriculture metrics spanning the years 1997-2021. The fields are State, Crop, District, Year, Season which are dimensions and Area, Area units, Production, Production units and Yields are measures.
Data Cleaning	Missing values: Isolated and deleted incomplete rows to prevent statistical skews.
Data Transformation	Calculated Fields: We created a new calculated field "Yield" which is the ratio of production per unit area. We also used filters for specifying a visual for a particular state and many other such filters.
Data Type Conversion	Validated that state, crop, district year, and season are classified as non-numeric strings. Numerical values like area, production are in million formats.
Column Splitting and Merging	Not needed.
Data Modeling	Not needed.

Save Processed Data	Exported the fully standardized dataset from raw system states into a permanently structured Excel file extension format, explicitly adjusting column widths to ensure complete field visibility before transforming to visualization platforms.
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